TEAM PROJECT PROPOSAL

CashFlowAI: Machine Learning for Small Business Cash-Flow Forecasting

Artificial Intelligence Group Project | 2026–2027 | Mekha Molly Raju, Mert Pehlivan, Rahul Ravi

1. Problem and initial research

Small businesses need to plan for periods when payments received may not cover outgoing payments. Forecasting future cash movements could help identify shortfalls earlier. Existing accounting products, such as Float and Xero, provide cash-flow planning tools, but our project will test whether machine-learning methods improve forecasts over simple historical baselines (Float, n.d.). Forecasting research recommends evaluating methods against clear benchmarks and using past observations to predict later periods (Hyndman and Athanasopoulos, 2021; Makridakis et al., 2022). The project is an educational prototype, not a replacement for accounting software.

2. Dataset and research question

We plan to use the LabX Synthetic Banking Dataset (Labx.ee, 2024). Its published description reports 151.1 million transactions across January 2024–December 2025, including approximately 22.2 million transactions for 20,000 small and medium-sized enterprise (SME) customers. Relevant fields include customer segment, booking date, euro amount and credit/debit indicator. We will filter SME records and calculate weekly cash inflows, outflows and net movement (credits minus debits). This is bank-transaction cash flow, not full accounting cash flow. As the data are synthetic, findings cannot establish performance on real businesses. Data access, reuse permissions and quality will be checked before processing; a smaller, reproducible SME subset will be used.

Research question: How accurately can machine-learning models forecast the next week's net bank-transaction cash flow for simulated small businesses compared with simple forecasting methods?

3. SMART objectives

O1. By week 2, verify dataset access and terms, document its structure and produce a reproducible SME sample with quality checks.

O2. By week 4, construct weekly inflow, outflow and net-flow series for a defined set of eligible SMEs, and document missing periods and unusual transactions.

O3. By week 6, implement at least two simple forecasting baselines and one machine-learning approach using only information available before each forecast.

O4. By week 8, compare models using rolling time-based evaluation and report MAE and RMSE, including results by business where feasible.

O5. By the portfolio deadline, deliver a working demonstration with forecast plots, a clear limitations statement and reproducible code and documentation.

4. Scope

In scope: SME transaction selection; data cleaning and weekly aggregation; exploratory analysis; baseline and machine-learning forecasting; temporal evaluation; a small local dashboard;

documentation and ethical review. Out of scope: predicting full accounting statements, connecting to live bank accounts, real-time deployment, lending decisions, customer profiling, and claims of accuracy on real-world SME data.

5. Methodology and team organisation

We will use Data Driven Scrum with a GitHub Projects Kanban board. Each one- to two-week iteration will follow Create → Observe → Analyze: implement a task, inspect its outputs and tests, then record findings and next steps. GitHub Issues will define acceptance criteria; branches and pull requests will support peer review. The team will hold weekly planning/check-in meetings, iteration reviews and short retrospectives, recording decisions and experiments. Student A leads data preparation and quality checks; Student B leads forecasting and evaluation; Student C leads the dashboard, integration and repository documentation. All three review results, research sources and final outputs. Roles may rotate to share learning.

6. Work packages and implementation plan

WP1 (weeks 1–2; all): literature review, dataset licence/access check, risk log, backlog and research design. Output: project plan and documented data decision.

WP2 (weeks 2–4; A lead): SME extraction, cleaning, aggregation and exploratory analysis. Output: reproducible data pipeline and plots.

WP3 (weeks 4–6; B lead): forecasting baselines, lagged features and candidate ML model. Output: versioned experiments.

WP4 (weeks 6–8; B lead, A review): rolling evaluation, leakage checks, error analysis and model selection. Output: comparison report.

WP5 (weeks 7–9; C lead): local dashboard, tests, README, portfolio evidence and presentation rehearsal. Output: demonstration and public repository.

7. Feasibility, risks and responsible use

The project will use free Python tools and local processing of a filtered dataset to limit computing costs. Main risks are inaccessible or restricted data, excessive volume, artificial patterns, sparse business histories, leakage and weak model results. We will check access early, use column selection and chunked processing, exclude businesses with insufficient history, freeze time-based test periods and retain baseline results even if ML performs worse. Synthetic records reduce direct privacy concerns but do not remove risks of misleading conclusions. We will document provenance, usage terms, bias, limitations and the need for human judgement before any financial use.

References

Float. (n.d.). Float cash flow forecasting software | xero, QBO & FreeAgent. [online] Available at: https://floatapp.com/ [Accessed 18 Oct. 2026].

Hyndman, R.J. and Athanasopoulos, G. (2021). Forecasting: Principles and practice. [online] otexts.com. OTexts. Available at: https://otexts.com/fpp3/ [Accessed 18 Oct. 2026].

Labx.ee. (2024). Synthetic banking dataset. [online] Available at: https://synthetic.labx.ee/ [Accessed 16 Oct. 2026].

Makridakis, S., Spiliotis, E. and Assimakopoulos, V. (2022). M5 accuracy competition: Results, findings, and conclusions. International Journal of Forecasting, 38(4). doi:10.1016/j.ijforecast.2021.11.013 [Accessed 18 Oct. 2026].